1 Overview and Descriptive Statistics

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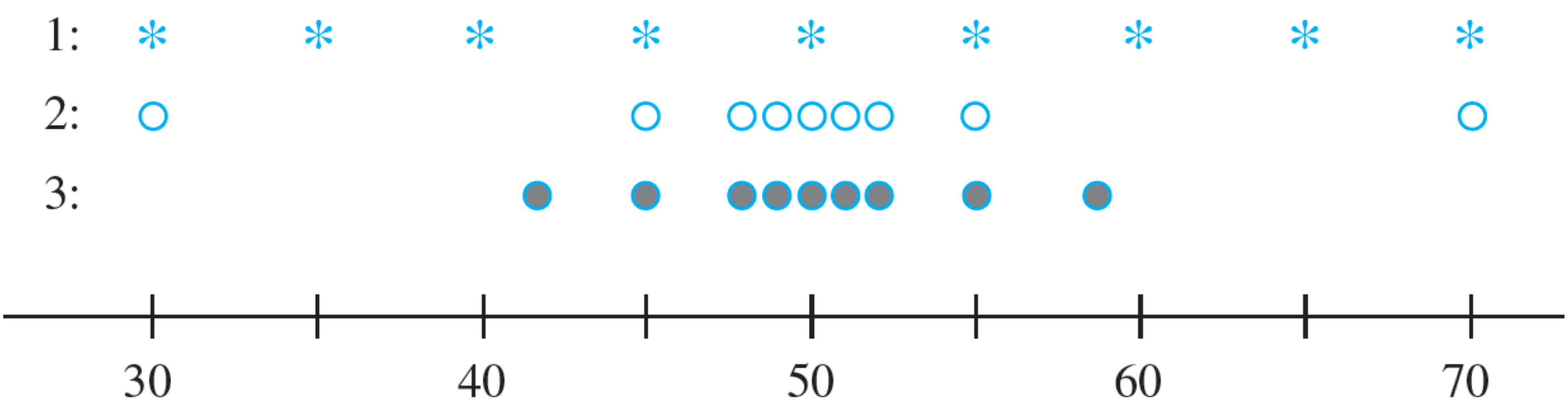
1.4 Measures of Variability

Measures of Variability

Reporting a measure of center gives only partial information about a data set or distribution. Different samples or populations may have identical measures of center yet differ from one another in other important ways.

The following Figure shows dotplots of three samples with the same mean and median, yet the extent of spread about the center is different for all three samples.

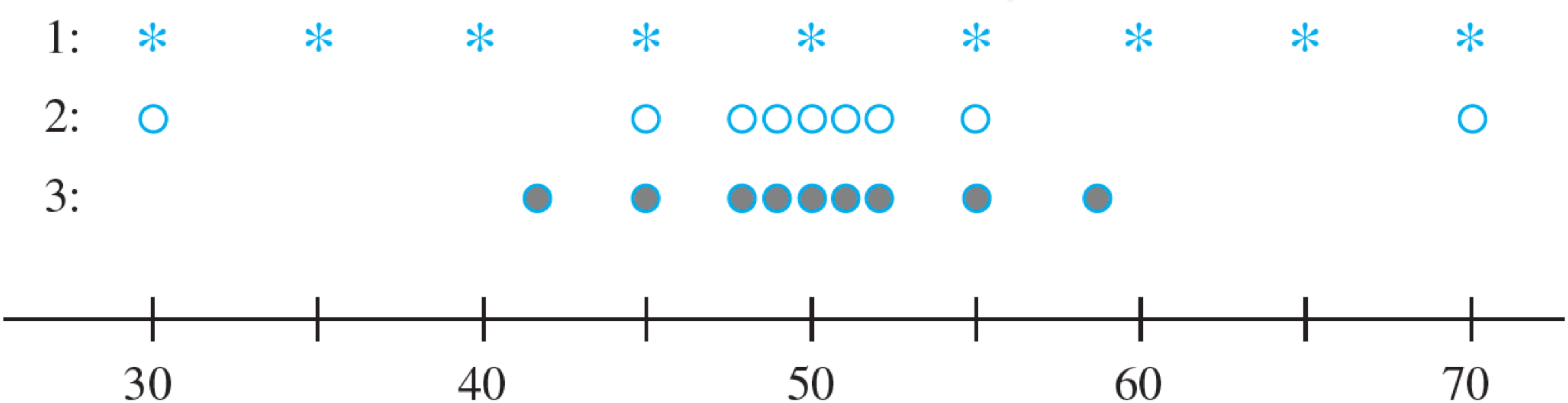


Samples with identical measures of center but different amounts of variability

Measures of Variability for Sample Data

Measures of Variability for Sample Data

The simplest measure of variability in a sample is the **range**, which is the difference between the largest and smallest sample values. The value of the range for sample 1 is much larger than it is for sample 3, reflecting more variability in the first sample than in the third.



Samples with identical measures of center but different amounts of variability

Measures of Variability for Sample Data

A defect of the range, though, is that it depends on only the two most extreme observations and disregards the positions of the remaining n – 2 values.

Samples 1 and 2 in the previous Figure have identical ranges, yet there is much less variability or dispersion in the second sample than in the first.

Our primary measures of variability involve the **deviations from the mean:**

$$x_1 - \bar{x}, x_2 - \bar{x}, \ldots, x_n - \bar{x}.$$

That is, the deviations from the mean are obtained by subtracting $\bar{x}$ from each of the n sample observations.

Measures of Variability for Sample Data

A deviation will be positive if the observation is larger than the mean (to the right of the mean on the measurement axis) and negative if the observation is smaller than the mean. If all the deviations are small in magnitude, then all x_is are close to the mean and there is little variability.

Alternatively, if some of the deviations are large in magnitude, then some x_is lie far from $\bar{x}$ suggesting a greater amount of variability.

A simple way to combine the deviations into a single quantity is to average them.

Measures of Variability for Sample Data

Unfortunately, this is a bad idea:

$$\text{sum of deviations} = \sum_{i=1}^{n} (x_i - \bar{x}) = 0$$

so that the average deviation is always zero. The verification uses several standard rules of summation and the fact that $\sum \bar{x} = \bar{x} + \bar{x} + \cdots + \bar{x} = n\bar{x}$:

$$\sum (x_i - \bar{x}) = \sum x_i - \sum \bar{x} = \sum x_i - n\bar{x} = \sum x_i - n\left(\frac{1}{n}\sum x_i\right) = 0$$

How can we prevent negative and positive deviations from counteracting one another when they are combined?

Measures of Variability for Sample Data

One possibility is to work with the absolute values of the deviations and calculate the average absolute deviation

$$\sum |x_i - \bar{x}|/n.$$

Because the absolute value operation leads to a number of theoretical difficulties, **consider instead the squared deviations**

$$(x_1 - \bar{x})^2, (x_2 - \bar{x})^2, \ldots, (x_n - \bar{x})^2.$$

Rather than use the average squared deviation $\sum(x_i - \bar{x})^2/n$, for several reasons we divide the sum of squared deviations by $n - 1$ rather than n.

Measures of Variability for Sample Data

Definition

The **sample variance**, denoted by s^2, is given by

$$s^2 = \frac{\sum (x_i - \bar{x})^2}{n-1} = \frac{S_{xx}}{n-1}$$

The **sample standard deviation**, denoted by s, is the (positive) square root of the variance:

$$s = \sqrt{s^2}$$

Note that s^2 and s are both nonnegative. The unit for s is the same as the unit for each of the x_is.

Example

Fuel Efficiency Data:

Consider the following

sample of n = 11 efficiencies for

the 2009 Ford Focus equipped with an automatic transmission (for this model, EPA reports an overall rating of

27 mpg–24 mpg for city driving and

33 mpg for highway driving):

Example

cont'd

Car	x_i	$x_i - \overline{x}$	$(x_i - \overline{x})^2$	
1	27.3	−5.96	35.522	
2	27.9	−5.36	28.730	
3	32.9	−0.36	0.130	
4	35.2	1.94	3.764	
5	44.9	11.64	135.490	
6	39.9	6.64	44.090	
7	30.0	−3.26	10.628	
8	29.7	−3.56	12.674	
9	28.5	−4.76	22.658	
10	32.0	−1.26	1.588	
11	37.6	4.34	18.836	
	$\Sigma x_i = 365.9$	$\Sigma(x_i - \overline{x}) = .04$	$\Sigma(x_i - \overline{x})^2 = 314.110$	$\overline{x} = 33.26$

Example

cont'd

Effects of rounding account for the sum of deviations not being exactly zero. The numerator of s^2 is S_{xx} = 314.106, from which

$$s^2 = \frac{S_{xx}}{n-1}$$

$$= \frac{314.106}{11-1}$$

$$= 31.41,$$

$$s = 5.60$$

The size of a representative deviation from the sample mean 33.26 is roughly 5.6 mpg.

A Computing Formula for s^2

A Computing Formula for s^2

There is an alternative formula for S_{xx} that avoids calculating the deviations.

$$S_{xx} = \sum (x_i - \overline{x})^2 = \sum x_i^2 - \frac{\left(\sum x_i\right)^2}{n}$$

Proof Because $\overline{x} = \Sigma x_i / n$, $n(\overline{x})^2 = \left(\Sigma x_i\right)^2 / n$. Then,

$$\begin{aligned} \sum (x_i - \overline{x})^2 &= \sum (x_i^2 - 2\overline{x} \cdot x_i + \overline{x}^2) = \sum x_i^2 - 2\overline{x} \sum x_i + \sum (\overline{x})^2 \\ &= \sum x_i^2 - 2\overline{x} \cdot n\overline{x} + n(\overline{x})^2 = \sum x_i^2 - n(\overline{x})^2 \end{aligned}$$

Example

Traumatic knee dislocation often requires surgery to repair ruptured ligaments. One measure of recovery is range of motion (measured as the angle formed when, starting with the leg straight, the knee is bent as far as possible).

The given data on postsurgical range of motion appeared in the article "Reconstruction of the Anterior and Posterior Cruciate Ligaments After Knee Dislocation" (*Amer. J. Sports Med.,* 1999: 189–197):

154 142 137 133 122 126 135 135 108 120 127 134 122

Example

cont'd

The sum of these 13 sample observations is $\sum x_i = 1695$, and the sum of their squares is

$$\sum x_i^2 = (154)^2 + (142)^2 + \cdots + (122)^2$$

$$= 222{,}581$$

Thus the numerator of the sample variance is

$$S_{xx} = \sum x_i^2 - [(\sum x_i)^2]/n$$

$$= 222{,}581 - (1695)^2/13$$

$$= 1579.0769$$

Example

cont'd

from which (n-1 = 12)

$$s^2 = 1579.0769/12$$

$$= 131.59$$

and

$$s = 11.47.$$

A Computing Formula for s^2

Proposition

Let $x_1, x_2, \ldots, x_n$ be a sample and c be any nonzero constant.

1. If $y_1 = x_1 + c, y_2 = x_2 + c, \ldots, y_n = x_n + c$, then $s_y^2 = s_x^2$, and
2. If $y_1 = cx_1, \ldots, y_n = cx_n$, then $s_y^2 = c^2 s_x^2$, $s_y = |c| s_x$

where s_x^2 is the sample variance of the x's

s_y^2 is the sample variance of the y's.

Boxplots

Boxplots

Stem-and-leaf displays and histograms convey rather general impressions about a data set, whereas a single summary such as the mean or standard deviation focuses on just one aspect of the data.

In recent years, a pictorial summary called a *boxplot* has been used successfully to describe several of a data set's most prominent features.

These features include (1) center, (2) spread, (3) the extent and nature of any departure from symmetry, and (4) identification of "outliers," observations that lie unusually far from the main body of the data.

Boxplots

Because even a single outlier can drastically affect the values of $\bar{x}$ and *s*, a boxplot is based on measures that are "resistant" to the presence of a few outliers—**the median** and a measure of variability called ***the fourth spread***.

Order the n observations from smallest to largest and separate the smallest half from the largest half; the median $\tilde{x}$ is included in both halves if n is odd. Then the **lower fourth** is the median of the smallest half and the **upper fourth** is the median of the largest half. A measure of spread that is resistant to outliers is the **fourth spread** f_s, given by

$$f_s = \text{upper fourth} - \text{lower fourth}$$

Boxplots

Roughly speaking, the fourth spread is unaffected by the positions of those observations in the smallest 25% or the largest 25% of the data. Hence it is resistant to outliers.

The simplest boxplot is based on **the following five-number** summary:

smallest x_i lower fourth median upper fourth largest x_i

Example

The accompanying data consists of observations on the time until failure (1000s of hours) for a sample of turbochargers from one type of engine (from **"The Beta Generalized Weibull Distribution: Properties and Applications,"** ***Reliability Engr. and System Safety*****, 2012: 5–15).**

1.6	2.0	2.6	3.0	3.5	3.9	4.5	4.6	4.8	5.0
5.1	5.3	5.4	5.6	5.8	6.0	6.0	6.1	6.3	6.5
6.5	6.7	7.0	7.1	7.3	7.3	7.3	7.7	7.7	7.8
7.9	8.0	8.1	8.3	8.4	8.4	8.5	8.7	8.8	9.0

The five-number summary is as follows.

smallest: 1.6 lower fourth: 5.05 median: 6.5 upper fourth: 7.85 largest: 9.0

Example

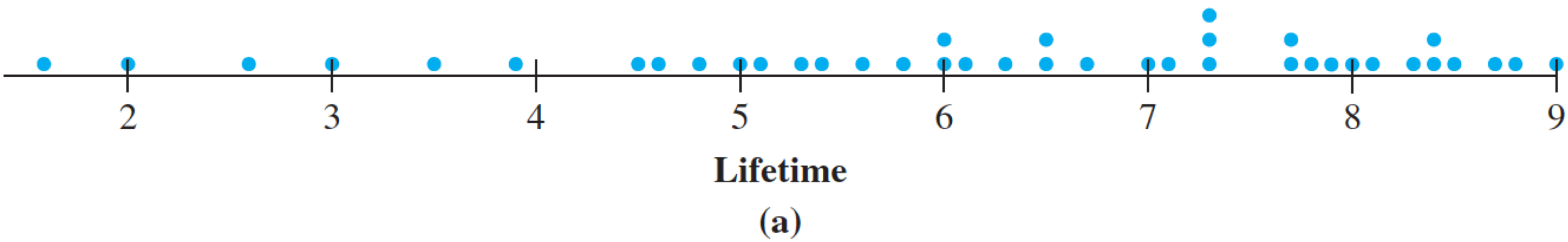
cont'd

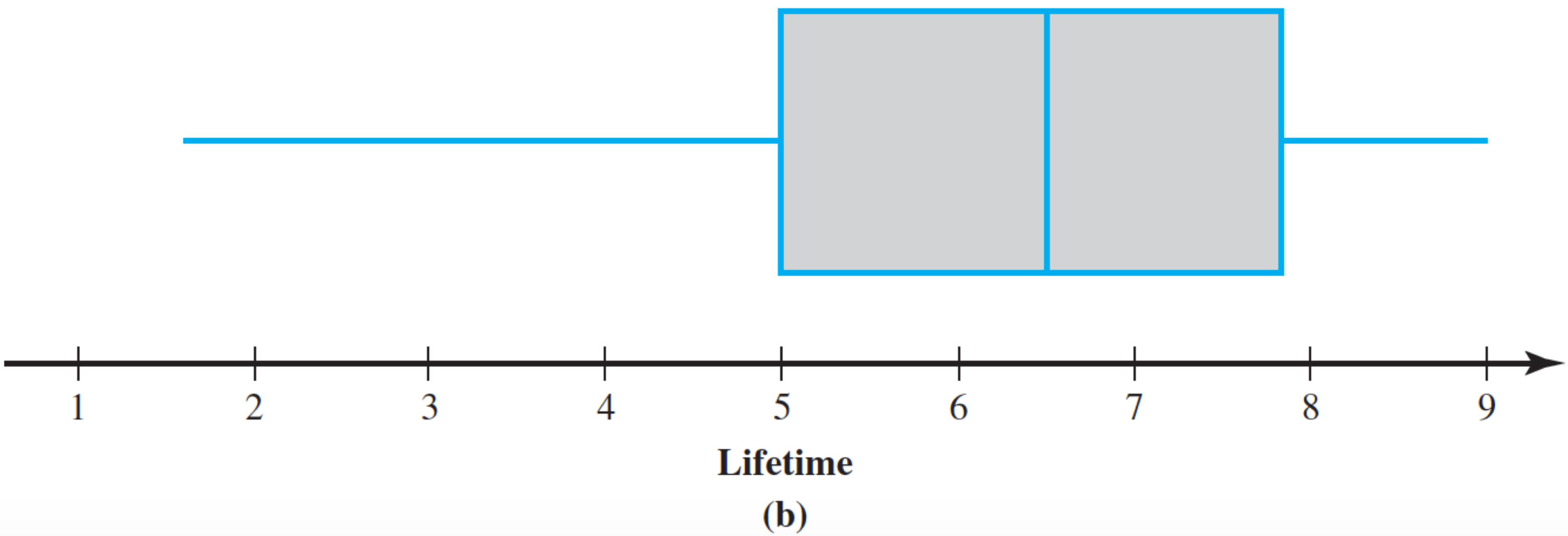
Variable	Count	Mean	SE Mean	StDev	Minimum
lifetime	40	6.253	0.309	1.956	1.600

Q1	Median	Q3	Maximum	IQR
5.025	6.500	7.875	9.000	2.850

Example

cont'd





(a) Dotplot and (b) Boxplot for the lifetime data

Boxplots That Show Outliers

Boxplots That Show Outliers

A boxplot can be embellished to indicate explicitly the presence of outliers. Many inferential procedures are based on the assumption that the population distribution is normal (a certain type of bell curve). Even a single extreme outlier in the sample warns the investigator that such procedures may be unreliable, and the presence of several mild outliers conveys the same message.

Definition

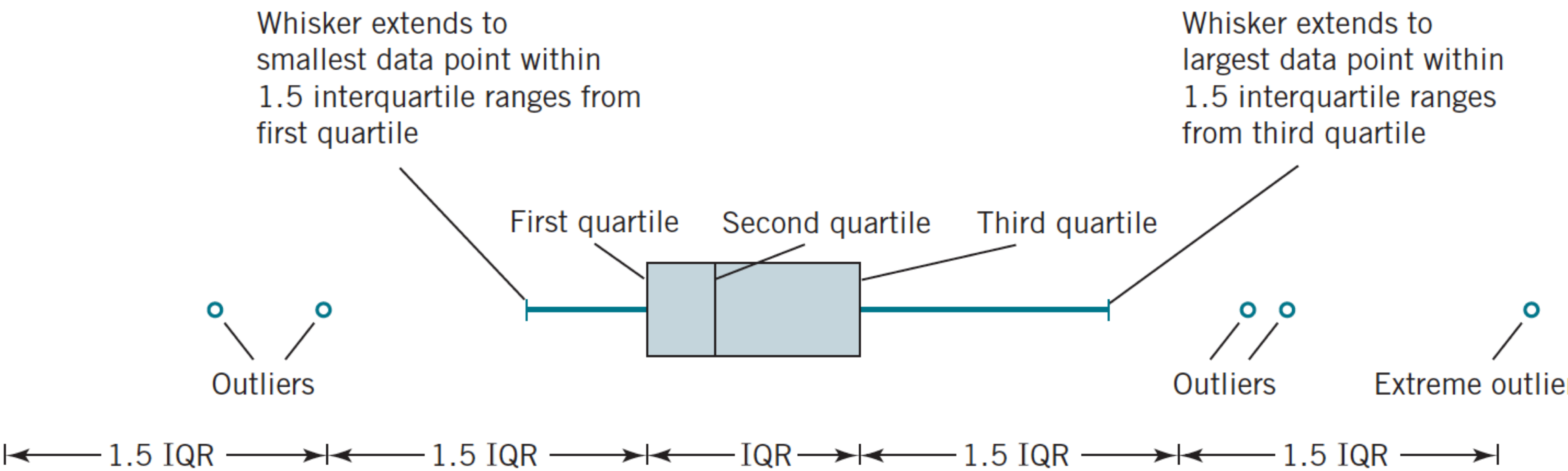
Any observation farther than $1.5f_s$ from the closest fourth is an **outlier**. An outlier is **extreme** if it is more than $3f_s$ from the nearest fourth, and it is **mild** otherwise.

Boxplots That Show Outliers

Let's now modify our previous construction of a boxplot by drawing a whisker out from each end of the box to the smallest and largest observations that are *not* outliers.

Each mild outlier is represented by a closed circle and each extreme outlier by an open circle. Some statistical computer packages do not distinguish between mild and extreme outliers.

Boxplots That Show Outliers



Example

The Clean Water Act and subsequent amendments require that all waters in the United States meet specific pollution reduction goals to ensure that water is "fishable and swimmable."

The article "Spurious Correlation in the USEPA Rating Curve Method for Estimating Pollutant Loads" (*J. of Environ. Engr.*, 2008: 610–618) investigated various techniques for estimating pollutant loads in watersheds; the authors "discuss the imperative need to use sound statistical methods" for this purpose.

Example

cont'd

Among the data considered is the following sample of TN (total nitrogen) loads (kg N/day) from a particular Chesapeake Bay location, displayed here in increasing order.

9.69	13.16	17.09	18.12	23.70	24.07	24.29	26.43
30.75	31.54	35.07	36.99	40.32	42.51	45.64	48.22
49.98	50.06	55.02	57.00	58.41	61.31	64.25	65.24
66.14	67.68	81.40	90.80	92.17	92.42	100.82	101.94
103.61	106.28	106.80	108.69	114.61	120.86	124.54	143.27
143.75	149.64	167.79	182.50	192.55	193.53	271.57	292.61
312.45	352.09	371.47	444.68	460.86	563.92	690.11	826.54
1529.35							

Example

cont'd

Relevant summary quantities are

$$\tilde{x} = 92.17 \qquad \text{lower } 4^{\text{th}} = 45.64 \qquad \text{upper } 4^{\text{th}} = 167.79$$

$$f_s = 122.15 \qquad 1.5f_s = 183.225 \qquad 3f_s = 366.45$$

Subtracting $1.5f_s$ from the lower 4th gives a negative number, and none of the observations are negative, so there are no outliers on the lower end of the data.

However,

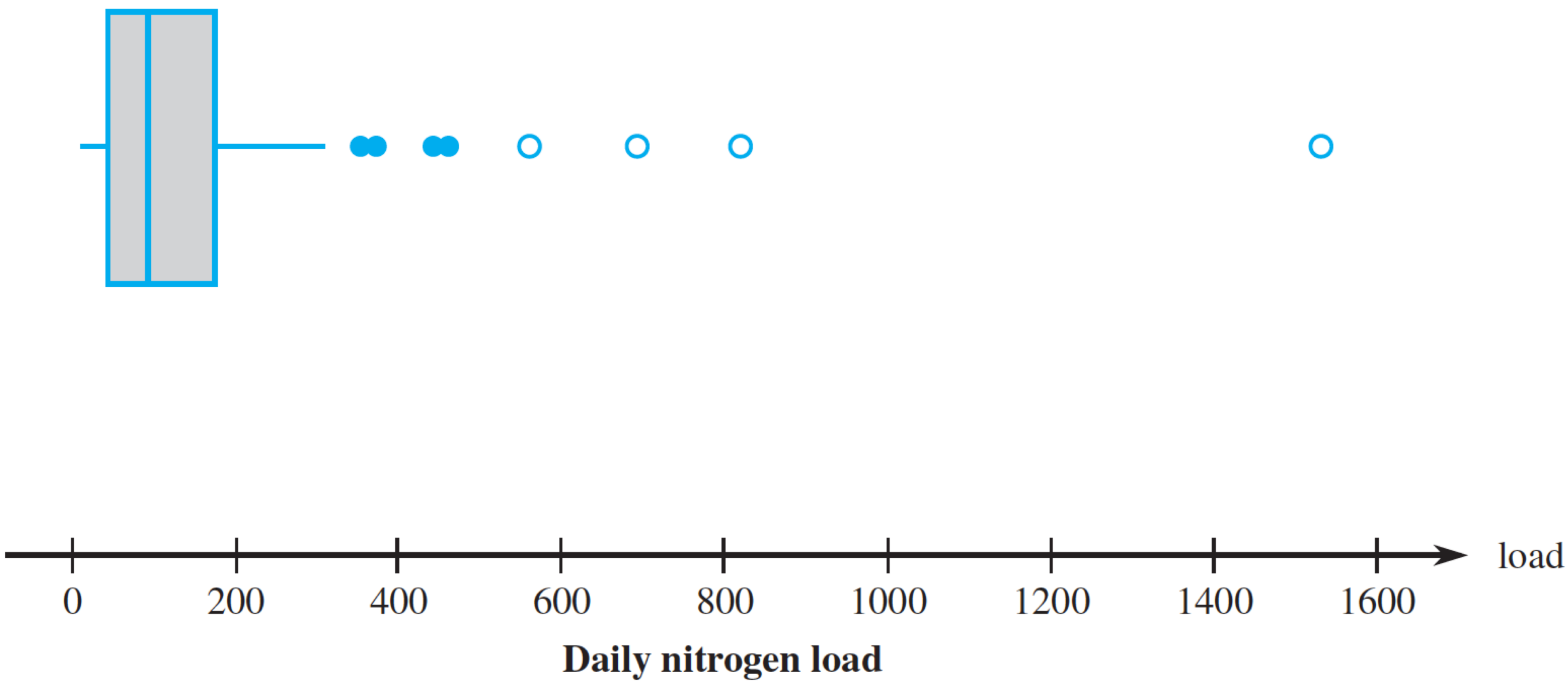
upper $4^{\text{th}} + 1.5f_s = 351.015$ upper $4^{\text{th}} + 3f_s = 534.24$

Thus the four largest observations—563.92, 690.11, 826.54, and 1529.35—are extreme outliers, and 352.09, 371.47, 444.68, and 460.86 are mild outliers.

Example

cont'd

The whiskers in the boxplot in the following Figure extend out to the smallest observation, 9.69, on the low end and 312.45, the largest observation that is not an outlier:



A boxplot of the nitrogen load data showing mild and extreme outliers

Comparative Boxplots

Comparative Boxplots

A comparative or side-by-side boxplot is a very effective way of revealing similarities and differences between two or more data sets consisting of observations on the same variable—fuel efficiency observations for four different types of automobiles, crop yields for three different varieties, and so on.

Example

In recent years, some evidence suggests that high indoor radon concentration may be linked to the development of childhood cancers, but many health professionals remain unconvinced.

A recent article ("Indoor Radon and Childhood Cancer," *The Lancet,* 1991: 1537–1538) presented the accompanying data on radon concentration (Bq/m3) in two different samples of houses. The first sample consisted of houses in which a child diagnosed with cancer had been residing.

Example

Houses in the second sample had no recorded cases of childhood cancer. The following Figure presents a stem-and-leaf display of the data.

1. Cancer		2. No cancer
9683795	0	95768397678993
86071815066815233150	1	12271713114
12302731	2	99494191
8349	3	839
5	4	
7	5	55
	6	
	7	
	8	5

HI: 210

Stem: Tens digit

Leaf: Ones digit

Stem-and-leaf display

Example

cont'd

Numerical summary quantities are as follows:

	$\overline{x}$	$\widetilde{x}$	s	f_s
Cancer	22.8	16.0	31.7	11.0
No cancer	19.2	12.0	17.0	18.0

The values of both the mean and median suggest that the cancer sample is centered somewhat to the right of the no-cancer sample on the measurement scale.

Example

cont'd

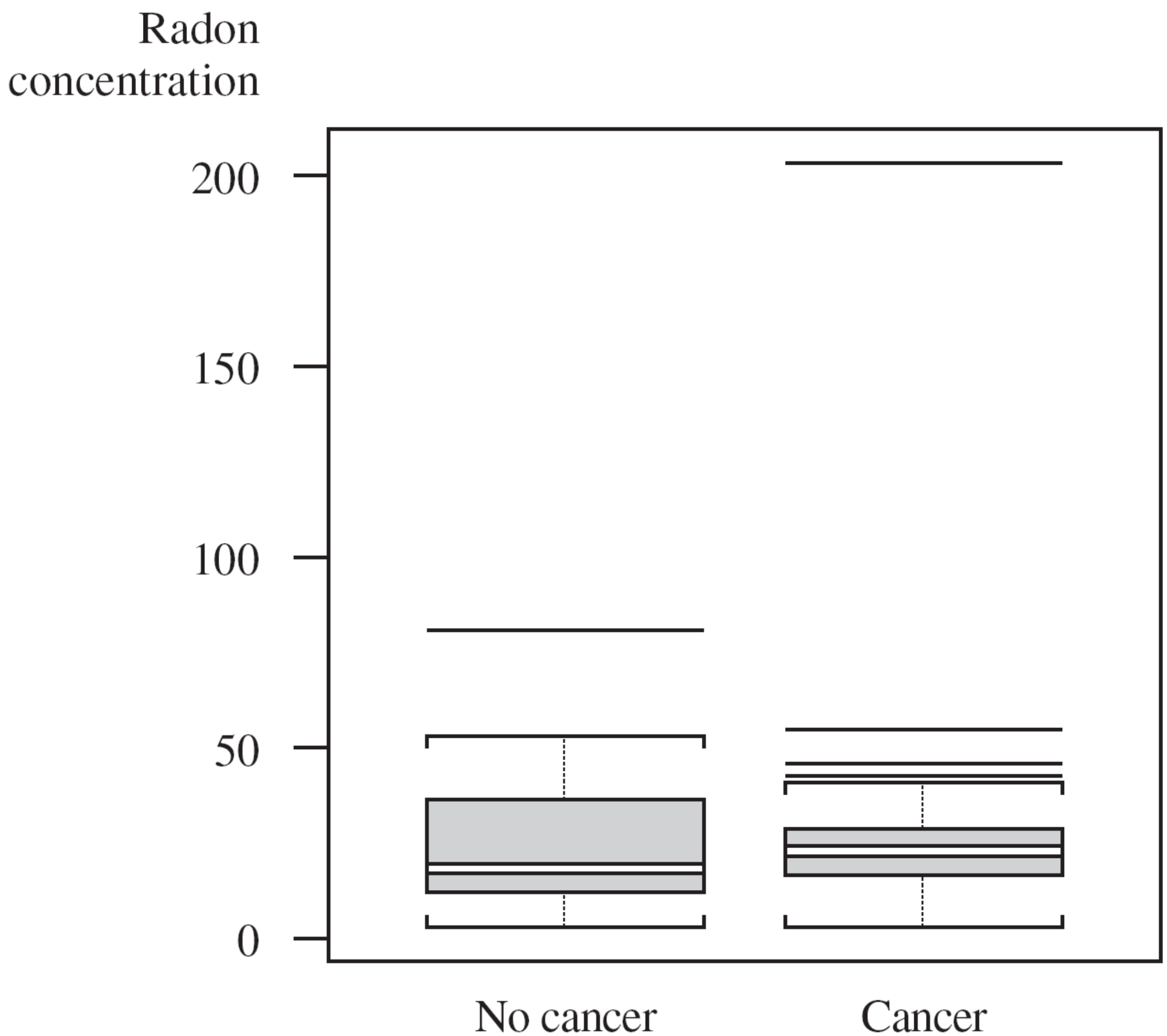
The mean, however, exaggerates the magnitude of this shift, largely because of the observation 210 in the cancer sample.

The values of *s* suggest more variability in the cancer sample than in the no-cancer sample, but this impression is contradicted by the fourth spreads. Again, the observation 210, an extreme outlier, is the culprit.

Example

cont'd

The Figure shows a comparative boxplot (using S-Plus).



A boxplot of the data

Example

cont'd

The no-cancer box is stretched out compared with the cancer box (f_s = 18 vs. f_s = 11), and the positions of the median lines in the two boxes show much more skewness in the middle half of the no-cancer sample than the cancer sample.

Outliers are represented by horizontal line segments, and there is no distinction between mild and extreme outliers.